

CONTACT INFORMATION	Department of Mathematics, UC San Diego, 9500 Gilman Drive, La Jolla, CA, USA, 92093 Office: HSS 4047	prangriz@ucsd.edu https://prangriz.github.io
EDUCATION	PhD in Mathematics. University of California, San Diego (Sep 2025 - Current) <i>Advisor:</i> Todd Kemp MMath in Statistics. University of Waterloo, (Sep 2023 - June 2025) <i>Advisor:</i> Aukosh Jagannath <i>Thesis:</i> High-dimensional scaling limits of online SGD in single-index models BSc in Physics. Sharif University of Technology (Sep 2018 - Feb 2023) <i>Minor:</i> Mathematics	
PUBLICATIONS & PREPRINTS	P. Rangriz, <i>Limit Theorems for Stochastic Gradient Descent in High-Dimensional Single-Layer Networks</i> , arXiv:2511.02258 [stat.ML], 2026. P. Rangriz, <i>High-Dimensional Scaling Limits of Online Stochastic Gradient Descent in Single-Index Models</i> , Master's Thesis, University of Waterloo, 2025.	
EMPLOYMENT	Graduate Research Assistant. University of Waterloo (Sep 2023 - Apr 2025) <i>Supervisor:</i> Aukosh Jagannath Worked on high-dimensional probability, scaling limits, random matrices, and mathematical foundations of machine learning. Summer Research Intern. EPFL (July 2022 - Sep 2022) <i>Supervisor:</i> Lenka Zdeborova Worked on statistical physics methods for machine learning and combinatorial optimization, e.g., mean-field theory, replica method, and message-passing algorithms.	
AWARDS & HONORS	First-Year Departmental Fellowship. UC San Diego (2025 - 2026) Graduate Research Studentship. University of Waterloo (2023 - 2025) International Master's Award of Excellence. University of Waterloo (2023) Master of Math Entrance Scholarship. University of Waterloo (2023) Summer@EPFL Fellowship. EPFL (2022) Silver Medal in the 30th Iran National Physics Olympiad (2017)	
TEACHING ASSISTANT EXPERIENCE	UC San Diego (2025-Present) PHYS 1CL: Waves, Optics & Modern Physics Lab (Summer 2026); MATH 193A: Actuarial Mathematics (Summer 2026); MATH 180A: Introduction to Probability (Spring 2026); MATH 10B: Calculus II (Winter 2026); MATH 11: Calculus-Based Probability and Statistics (Fall 2025) University of Waterloo (2023-2025) STAT 333: Stochastic Processes I (Winter 2025, Winter 2024); STAT 433: Stochastic Processes II (Fall 2024); STAT 330: Mathematical Statistics (Fall 2023, Spring 2024); STAT 231: Probability (Spring 2024); STAT 230: Statistics (Spring 2024); STAT 202: Introductory Statistics for Scientists (Fall 2023) Sharif University of Technology (2019-2022) Advanced Statistical Mechanics (Fall 2022, Fall 2021, Spring 2021); Statistical Mechanics (Spring 2021); Thermodynamics (Fall 2020); General Physics III (Fall 2019); Fundamentals of C Programming (Fall 2019)	